

Arduino Code

```
#include <SoftwareSerial.h>
```

```
#include <Servo.h>
```

```
Servo servo01;
```

```
Servo servo02;
```

```
Servo servo03;
```

```
Servo servo06;
```

```
#define relay1 6
```

```
#define relay2 7
```

```
#define relay3 8
```

```
#define relay4 9
```

```
SoftwareSerial Bluetooth(0, 1); // Arduino(RX, TX) - HC-05 Bluetooth (TX, RX)
```

```
int wheelSpeed = 1500;
```

```
int lbw[50], lfw[50], rbw[50], rfw[50]; // arrays for storing positions/steps
```

```
int servo1Pos, servo2Pos, servo3Pos, servo4Pos, servo5Pos, servo6Pos; // current position
```

```
int servo1PPos, servo2PPos, servo3PPos, servo4PPos, servo5PPos, servo6PPos; // previous position
```

```
int servo01SP[50], servo02SP[50], servo03SP[50], servo04SP[50], servo05SP[50], servo06SP[50]; //  
for storing positions/steps
```

```
int speedDelay = 20;
```

```
int index = 0;
```

```
int dataIn;
```

```
int m = 0;
```

```
void setup() {

    pinMode(relay1, OUTPUT);
    pinMode(relay2, OUTPUT);
    pinMode(relay3, OUTPUT);
    pinMode(relay4, OUTPUT);

    digitalWrite(relay1, LOW);
    digitalWrite(relay2, LOW);
    digitalWrite(relay3, LOW);
    digitalWrite(relay4, LOW);

    // Set initial seed values for the steppers
    servo01.attach(2);
    servo02.attach(3);
    servo03.attach(4);
    servo06.attach(5);
    Bluetooth.begin(9600); // Default baud rate of the Bluetooth module
    Bluetooth.setTimeout(5);
    delay(20);
    Serial.begin(38400);
    // Move robot arm to initial position
    servo1PPos = 90;
    servo01.write(servo1PPos);
    servo2PPos = 100;
    servo02.write(servo2PPos);
    servo3PPos = 120;
    servo03.write(servo3PPos);
    servo6PPos = 110;
    servo06.write(servo6PPos);
}
```

```
void loop() {  
  // Check for incoming data  
  if (Bluetooth.available() > 0) {  
    dataIn = Bluetooth.read(); // Read the data  
    Serial.println(dataIn);  
  
    if (dataIn == 0) {  
      m = 0;  
    }  
    if (dataIn == 1) {  
      m = 1;  
    }  
    if (dataIn == 2) {  
      m = 2;  
    }  
    if (dataIn == 3) {  
      m = 3;  
    }  
    if (dataIn == 4) {  
      m = 4;  
    }  
    if (dataIn == 5) {  
      m = 5;  
    }  
    if (dataIn == 6) {  
      m = 6;  
    }  
    if (dataIn == 7) {  
      m = 7;  
    }  
  }  
}
```

```
if (dataIn == 8) {  
    m = 8;  
}  
if (dataIn == 9) {  
    m = 9;  
}  
if (dataIn == 10) {  
    m = 10;  
}  
if (dataIn == 11) {  
    m = 11;  
}  
if (dataIn == 12) {  
    m = 12;  
}  
if (dataIn == 14) {  
    m = 14;  
}  
if (dataIn == 16) {  
    m = 16;  
}  
if (dataIn == 17) {  
    m = 17;  
}  
if (dataIn == 18) {  
    m = 18;  
}  
if (dataIn == 19) {  
    m = 19;  
}  
if (dataIn == 20) {
```

```
    m = 20;
}
if (dataIn == 21) {
    m = 21;
}
if (dataIn == 22) {
    m = 22;
}
if (dataIn == 23) {
    m = 23;
}
if (dataIn == 24) {
    m = 24;
}
if (dataIn == 25) {
    m = 25;
}
if (dataIn == 26) {
    m = 26;
}
if (dataIn == 27) {
    m = 27;
}
```

```
// Move the Mecanum wheels platform
```

```
if (m == 2) {
    moveForward();
}
if (m == 7) {
    moveBackward();
}
```

```
if (m == 9) {  
    rotateLeft();  
}  
  
if (m == 10) {  
    rotateRight();  
}  
  
if (m == 0) {  
    stopMoving();  
}  
  
// Mecanum wheels speed  
if (dataIn > 30 & dataIn < 100) {  
    wheelSpeed = dataIn * 20;  
}  
  
// Move robot arm  
// Move servo 1 in positive direction  
while (m == 16) {  
    if (Bluetooth.available() > 0) {  
        m = Bluetooth.read();  
    }  
    servo01.write(servo1PPos);  
    servo1PPos++;  
    delay(speedDelay);  
}  
  
// Move servo 1 in negative direction  
while (m == 17) {  
    if (Bluetooth.available() > 0) {  
        m = Bluetooth.read();  
    }  
}
```

```
servo01.write(servo1PPos);  
servo1PPos--;  
delay(speedDelay);  
}  
  
// Move servo 2  
while (m == 19) {  
    if (Bluetooth.available() > 0) {  
        m = Bluetooth.read();  
    }  
    servo02.write(servo2PPos);  
    servo2PPos++;  
    delay(speedDelay);  
}  
  
while (m == 18) {  
    if (Bluetooth.available() > 0) {  
        m = Bluetooth.read();  
    }  
    servo02.write(servo2PPos);  
    servo2PPos--;  
    delay(speedDelay);  
}  
  
// Move servo 3  
while (m == 20) {  
    if (Bluetooth.available() > 0) {  
        m = Bluetooth.read();  
    }  
    servo03.write(servo3PPos);  
    servo3PPos++;  
    delay(speedDelay);  
}  
  
while (m == 21) {
```

```
if (Bluetooth.available() > 0) {  
    m = Bluetooth.read();  
}  
servo03.write(servo3PPos);  
servo3PPos--;  
delay(speedDelay);  
}
```

```
// Move servo 6
```

```
while (m == 26) {  
    if (Bluetooth.available() > 0) {  
        m = Bluetooth.read();  
    }  
    servo06.write(servo6PPos);  
    servo6PPos++;  
    delay(speedDelay);  
}
```

```
while (m == 27) {  
    if (Bluetooth.available() > 0) {  
        m = Bluetooth.read();  
    }  
    servo06.write(servo6PPos);  
    servo6PPos--;  
    delay(speedDelay);  
}
```

```
// If arm speed slider is changed
```

```
if (dataIn > 101 & dataIn < 250) {  
    speedDelay = dataIn / 10; // Change servo speed (delay time)  
}
```



```

// If button "RUN" is pressed
if (m == 14) {
    runSteps();

    // If button "RESET" is pressed
    if (dataIn != 14) {
        stopMoving();

        memset(lbw, 0, sizeof(lbw)); // Clear the array data to 0
        memset(lfw, 0, sizeof(lfw));
        memset(rbw, 0, sizeof(rbw));
        memset(rfw, 0, sizeof(rfw));
        memset(servo01SP, 0, sizeof(servo01SP)); // Clear the array data to 0
        memset(servo02SP, 0, sizeof(servo02SP));
        memset(servo03SP, 0, sizeof(servo03SP));
        memset(servo06SP, 0, sizeof(servo06SP));
        index = 0; // Index to 0
    }
}

}

void moveForward() {
    digitalWrite(relay1, HIGH);
    digitalWrite(relay2, LOW);
    digitalWrite(relay3, HIGH);
    digitalWrite(relay4, LOW);
}

void moveBackward() {
    digitalWrite(relay1, LOW);

```

```
    digitalWrite(relay2, HIGH);
    digitalWrite(relay3, LOW);
    digitalWrite(relay4, HIGH);
}

void rotateLeft() {
    digitalWrite(relay1, HIGH);
    digitalWrite(relay2, LOW);
    digitalWrite(relay3, LOW);
    digitalWrite(relay4, HIGH);
}

void rotateRight() {
    digitalWrite(relay1, LOW);
    digitalWrite(relay2, HIGH);
    digitalWrite(relay3, HIGH);
    digitalWrite(relay4, LOW);
}

void stopMoving() {
    digitalWrite(relay1, LOW);
    digitalWrite(relay2, LOW);
    digitalWrite(relay3, LOW);
    digitalWrite(relay4, LOW);
}

// Automatic mode custom function - run the saved steps
void runSteps() {

}
```